I Putu Romyadhy Maha Putra

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Education

Universitas Pendidikan Ganehsa Singaraja, Bali
Bachelor of Science in Computer Science *August 2022 – Present*

- Select coursework: Networks, Machine Learning, Algorithms, Data Structures, OOP, IoT, Web Dev, Database

Technical Skills

Programming Languages: C/C++, HTML/CSS, JavaScript, Python, SQL, PHP
Software/Tools: Git, Vim, CMake, Linux; Docker, MySQL, Laravel, React.js, Vue.js, Next.js; Microsoft Office

Work Experience

Living Lab FTK Undiksha X Dago Engineering Singaraja, Bali
Student Researcher *February 2025 – Present*

- Build **IoT** system that can monitors the water quality using multiple parameter sensors, integrating IoT, distributing data via MQTT to a Raspberry Pi, and Firebase
- Built a Telegram bot to monitor sensor reading and manually control hydroponics relay and pump
- Collaborated with a team to write a research paper based on the project, submitted to GEMASTIK 2025

RSU Kertha Usada Singaraja, Bali
IT Support *August 2025 – December 2025*

- Build a **Helpdesk IT Support system**, that can report and store all the issue and report in several fields at hospital
- Presented and proposed new system that can record all problems and reports from several areas in the hospital updated real-time like via website using Laravel

Projects

Monitoring and Automations Hydroponics (Research Team) *February 2025 – Present*

- Building **IoT** for monitoring and doing some automations using machine learning with those data
- Researcher project in living lab that used all the tech fields in hydroponics things
- Making distributions data sensor, stored on RASPI, notifications on telegram bot
- Collaborate with the team and make a paper, and submitting this project into GEMASTIK

SamuBrawl Game (Solo Project) *May 2026 – August 2026*

- First game project built independently in C++ with **RAYLIB**, self-initiated to learn game development
- Built a 2D samurai fighting **game** inspired by **BRAWLHALLA**, with gameplay logic and controls implemented from scratch
- Released and published on **itch.io**

Helpdesk IT Support System (Solo Project) *August 2025 – December 2025*

- Build a system that can stored all the problem and report at **RSU Kertha Usada**, using Laravel
- Helped to making feature that can **record all the issue** in daily and making an **automatic report** in daily, monthly, until annual report

Semut Dry Tech – BALIGIVATION 2025 (Team Project) *August 2025 – December 2025*

- Team member on a project led by a faculty advisor, developing an ultrasonic drying oven for “gula semut” (palm sugar)
- Designed the hardware system's temperature and humidity sensing components
- Developed a mobile application for real-time monitoring

Ganesha Homecare (Team Project) *October 2024 – December 2024*

- Group Project about homecare at health center, help to maintenance website that used CodeIgniter
- Build an API for history and transactions feature, tested using postman. Deploying and config at health center in Badung, Bali
- Presented and joint discussion regarding the results of the website at the health center, and helping to create documentation for the website

Publications, Competition, Awards

- Submitted a research paper on IoT-based hydroponics monitoring and automation to **GEMASTIK**, as part of the Living Lab research project
- Currently adapting undergraduate thesis, *"IoT-Based Monitoring and Automation of Hydroponics Using Decision Tree for Water Quality and Nutrient Distribution,"* into a journal paper on **JITET**
- **2nd Place**, **BALIGIVATION** 2025 Bank Indonesia Digital Innovation Competition, Denpasar, Bali “***Semut Dry Tech***”